Practical Reconstruction

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Key Concepts

1) Ideal reconstruction involves applying a brick-wall low pass filter to an impulse-sampled signal.

    a) An impulse-sampled signal is a mathematical concept that cannot be physically generated.

    b) Ideal, brick wall low pass analog filters cannot be built.

2) Practical reconstruction uses a device called a zero-order hold to produce a continuous-time signal with amplitude proportional to the value of the discrete-time signal in each sampling interval.

a) The zero-order hold is represented mathematically as a weighted sum of time shifted rectangular pulses.

b) A mathematical model for the zero-order hold output is the convolution between a rectangular pulse and the impulse-sampled signal.

c) The Fourier transform (FT) of the zero-order hold output is the product of the FT of the rectangular pulse and the FT of the impulse-sampled signal.

3) The zero-order hold introduces three distortions to the reconstructed signal.

a) A time delay of one-half the rectangular pulse width.

b) Curvature in the main lobe of the FT of the rectangular pulse distorts the signal FT.

c) Images, or distorted replicates, of the signal FT remain at integer multiples of the sampling frequency due to the side lobes of the rectangular pulse FT.

d) The images and main lobe distortion are a consequence of the stair step nature of the zero-order hold output.

4) The output of the zero-order hold is passed through a low pass anti-imaging filter to eliminate (or greatly reduce) the images at multiples of the sampling frequency.

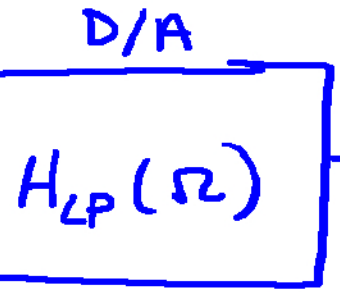
5) An anti-imaging filter can also reduce the main lobe distortion.

6) The anti-imaging filter is easier to build when the sampling frequency is much greater than twice the highest frequency in the signal to be reconstructed.

Video Screenshots

Practical Reconstruction

Ideal -

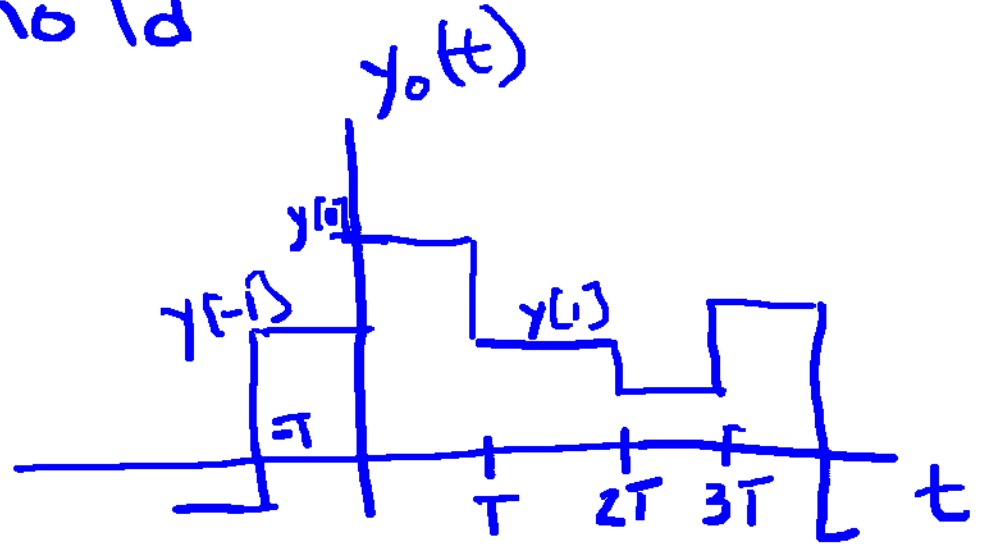


$y_s(t) = \sum_{n=-\infty}^{\infty} y[n]\delta(t-nT)$

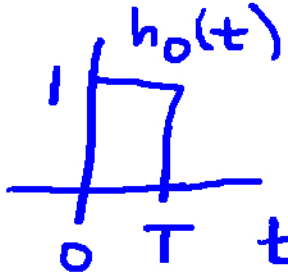
$$y(t) = T\sum_{n=-\infty}^{\infty} y[n]\,\frac{\sin\left(\frac{\Omega_s}{2}(t-nT)\right)}{\pi(t-nT)}$$

Practical – zero-order hold

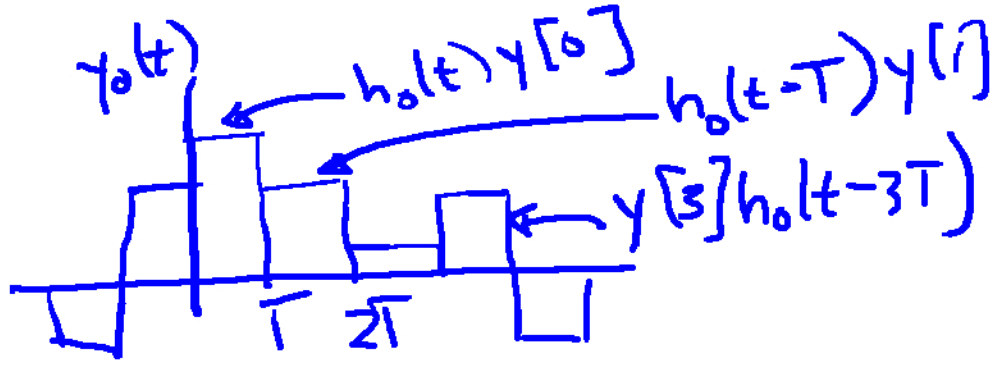
constant voltage of value y[n] is held for T sec



Let $h_0(t) = \begin{cases} 1 & 0 \le t \le T \\ 0 & \text{otherwise} \end{cases}$



$$y_0(t) = \sum_{n=-\infty}^{\infty} y[n]\, h_0(t-nT)$$



$$y_0(t) = \sum_{n=-\infty}^{\infty} y[n]\,\delta(t-nT) * h_0(t)$$

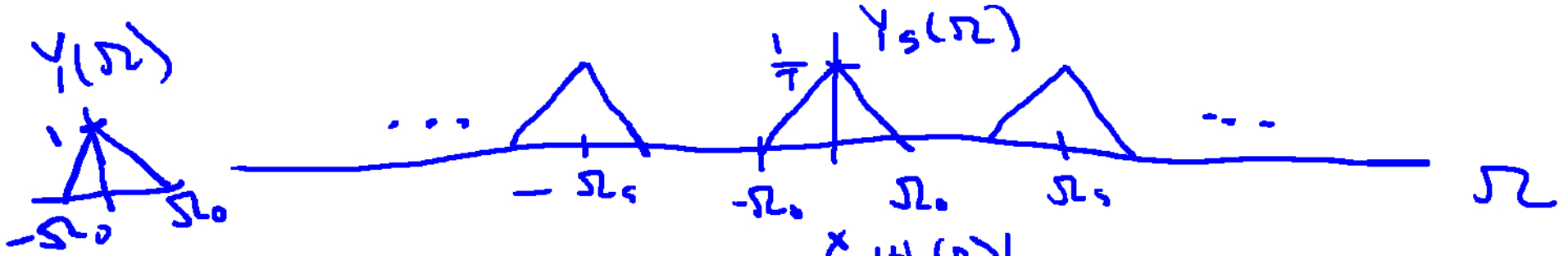
$$= y_s(t) * h_0(t)$$

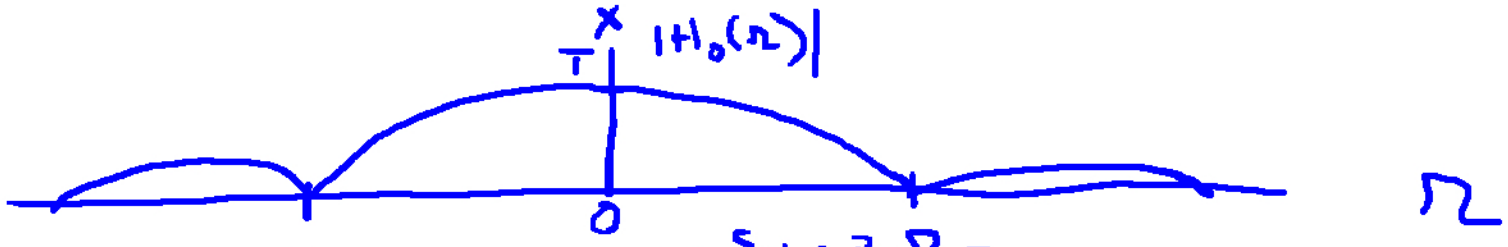
$$\Downarrow$$

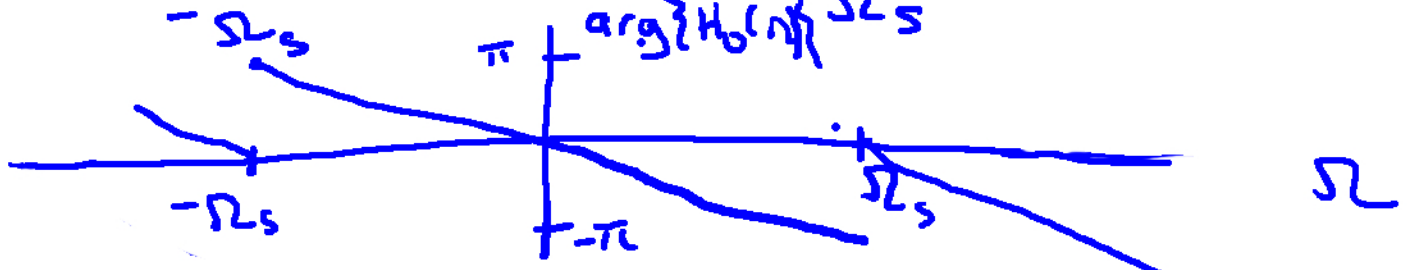
$$Y_0(\Omega) = Y_s(\Omega)\, H_0(\Omega)$$

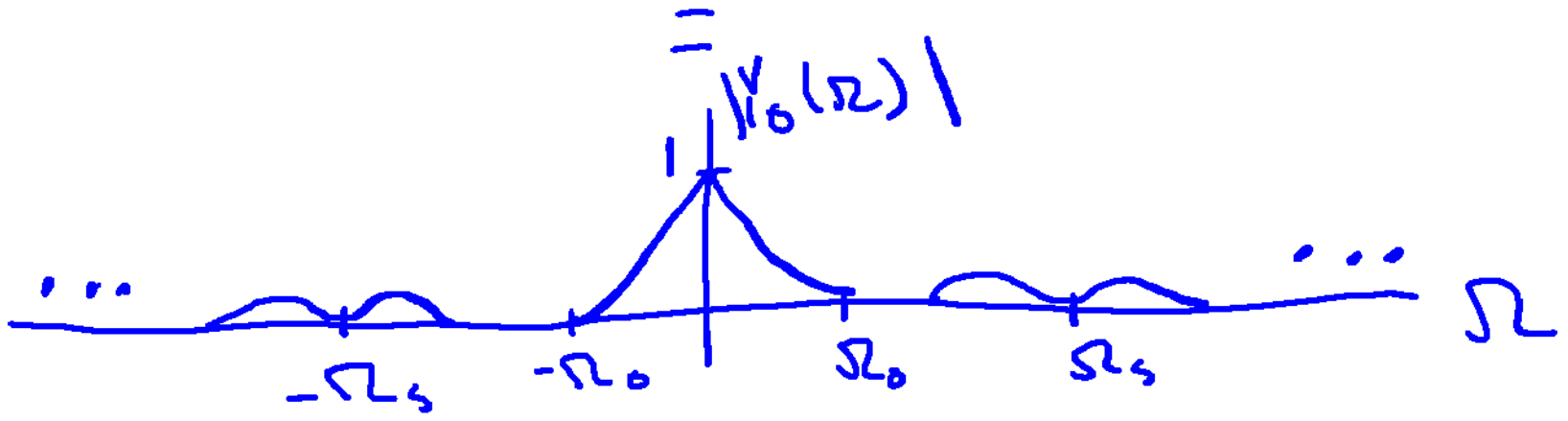
$$Y_s(\Omega) = \frac{1}{T}\sum_{l=-\infty}^{\infty} Y(\Omega - l\Omega_s)$$

$$H_0(\Omega) = 2e^{-j\Omega T/2}\,\frac{\sin\left(\frac{\Omega}{2}T\right)}{\Omega}$$


$Y_1(\Omega)$
1
$-\Omega_0$
Ω_0
$Y_s(\Omega)$
$\frac{1}{T}$
$-\Omega_s$
$-\Omega_0$
Ω_0
Ω_s
Ω


$|H_0(\Omega)|$
T
0
Ω


$-\Omega_s$
$\arg\{H_0(\Omega)\}$
π
Ω_s
$-\Omega_s$
$-\pi$
Ω


$|V_0(\Omega)|$
1
$-\Omega_s$
$-\Omega_0$
Ω_0
Ω_s
Ω

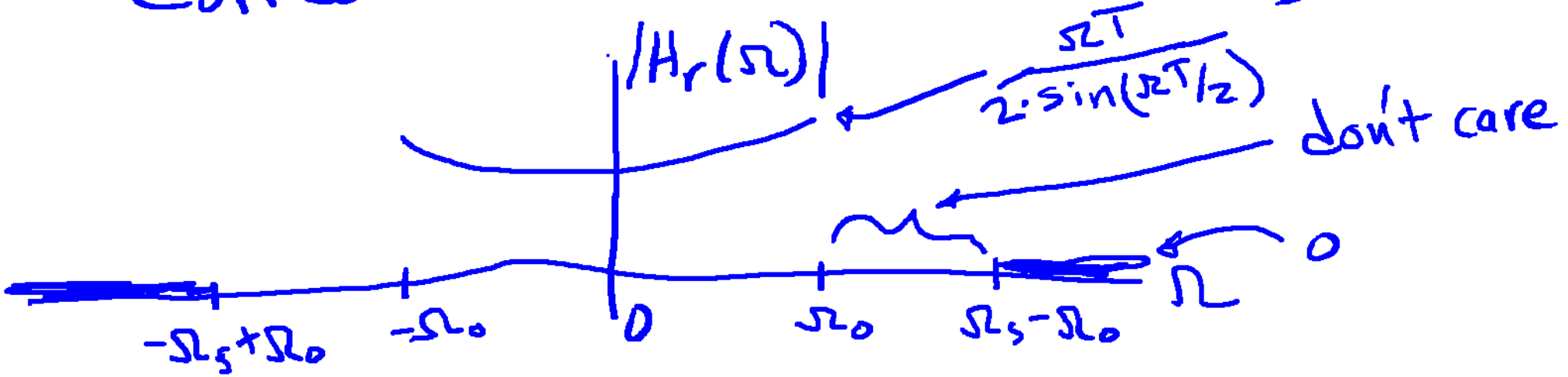
Effects of Zero-Order Hold

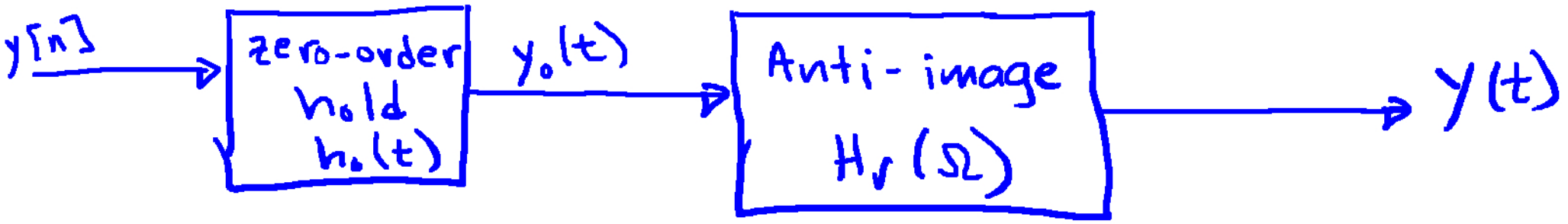
1) Linear phase shift $\Leftrightarrow$ time delay of $T/2$

2) Distortion between $-\Omega_0$ and Ω_0
(mainlobe of $H_0(\Omega)$)

3) Distorted/attenuated replicates of $Y(\Omega)$ at $k\Omega_s$ $(k \neq 0)$

Correct 2+3 with antiimaging filter





Goal: $|H_r(\Omega) H_0(\Omega)| = \begin{cases} T & |\Omega| < \Omega_0 \\ 0 & |\Omega| > \Omega_s - \Omega_0 \end{cases}$

$H_r(\Omega)$ is easier to build for $\Omega_s \gg \Omega_0$

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